

Shreshta Anupindi

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science in Aerospace Engineering

Expected Graduation: May 2027

- Minors: Applied Artificial Intelligence and Non-Destructive Evaluation
- Relevant Coursework: Aerodynamics, Compressible Flow, Flight Dynamics and Control, Aircraft Performance, Mechanics of Materials, Flight Structures, Astrodynamics, Computational Techniques for Aerospace Design

EXPERIENCE

LAUNCH-SPACE Research Intern

May 2026 - Aug 2026

NASA Iowa Space Grant Consortium, Dr. Xia

Ames, IA

- Conducted LAMMPS molecular dynamics simulations of grafted crosslinked polymer networks under tensile deformation to evaluate material mechanical response.
- Analyzed 4 material parameters including chain stiffness, interaction strength, grafting, and crosslinking to study their effects on deformation behavior and mechanical properties.

Open UAS Research Assistant

Jun 2024 - Present

Iowa State University, Dr. Rozier

Ames, IA

- Lead UAV electrical integration across 3 major system areas including power distribution, wiring harnesses, and avionics connections to support safe and reliable onboard operation.
- Serve as Flight Test Director by planning and executing UAV flight tests, coordinating ground crew operations, and verifying aircraft performance against safety and test objectives.

Undergraduate Research Assistant

Jan 2026 – Present

Iowa State University, Dr. He

Ames, IA

- Perform aerodynamic analysis of airfoils and wings using OpenFOAM-based simulations within DAFoam, evaluating lift, drag, and pressure distributions.
- Evaluated 3 aerodynamic design variables including angle of attack, wing twist, and airfoil geometry using adjoint-based optimization to improve aerodynamic efficiency.

PROJECTS

Wing Morphing Project (Personal Project) — SolidWorks

Jul 2025 - Present

- Develop parametric SolidWorks models of morphing wingtip concepts to investigate lift, drag, and aerodynamic efficiency across changing geometries.
- Evaluate structural feasibility, manufacturability, and mechanism complexity while balancing aerodynamic performance with practical aircraft design constraints.

Design Build and Fly (DBF) Competition

Aug 2023 - Apr 2024

- Coordinated operations, manufacturing, and technical reporting across interdisciplinary subteams, contributing to a 45th-place finish among 118 international teams.
- Facilitated communication between design, aerodynamics, and manufacturing teams to support aircraft integration, build progress, and competition readiness.

Aerospace Material Defect Detection

Sept 2024 - Nov 2024

- Developed a convolutional neural network (CNN) to detect defects in aerospace composite materials using image data for non-destructive evaluation applications.
- Preprocessed and augmented image datasets to improve model robustness and defect recognition across varying material conditions.

LEADERSHIP & ACTIVITIES

Indian Students Association, President (2026 - Present)

AIAA, Sponsorship Chair (2024 - Present)

TECHNICAL SKILLS

Engineering Design and Simulation: SolidWorks, Fusion 360, AutoCAD, OnShape, KeyShot, XFLR5, MATLAB, Simulink, LAMMPS

Programming, Testing and Tools: Python, VS Code, Wind Tunnel Testing, Sensor Integration, Data Analysis, Experimental Setup, Soldering, Wiring, Microsoft Excel, Google Workspace